Hurum Maksora Tohfa

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Education

University of Washington
PhD in Astrophysics — August 2023 - Present
MS in Astrophysics (GPA 3.9) — August 2023 - 2025

Bryn Mawr College
Magna Cum Laude (GPA 3.8)
Hons in Physics and Mathematics with concentration in scientific computing — September 2018 - May 2022
(Honors Thesis: CMB Spectral distortion due to variation in the fine structure constant)

Technical Skills

Programming Languages: Python, Julia, R, Java, Mathematica

Machine Learning: Neural Networks, Domain Adaptation, Deep Learning Frameworks

Data Analysis: Large-scale dataset processing, Bayesian statistics, Monte Carlo simulation, Time series analysis

High-Performance Computing: Parallel processing, GPU acceleration, Simulation frameworks (MP-Gadget, Batoid)

Mathematical Methods: Statistical modeling, Optimization, Numerical methods, Signal processing

Publications

- Tohfa, H.; et al.(2026). "Two bits about lossy compression: On the limits of compression in cosmology."(In review, accepted to Machine Learning and the Physical Sciences workshop, preprint version on ArXiv)
- Tohfa, H.; et al.(2026)."FOTO: Figure-Level Semantic Search For Astronomy."(Accepted to The Astrophysical Journal, Machine Learning and the Physical Sciences workshop, preprint version on ArXiv)
- Tohfa, H.; et al. (2026)."An emulator for the ionizing photon mean free path in ultra-high resolution simulations: the implications of mean free path measurements for the reionization history.". *Journal of Cosmology and Astroparticle Physics, JCAP06(2026)094, DOI 10.1088/1475-7516/2026/06/094*.
- Tohfa, H.; et al. (2023)."Forecast Cosmological Constraints with the 1D Wavelet Scattering Transform and the Lyman-α forest.". *Physical Review Letters (Vol. 132, No. 23). doi: 10.1103/PhysRevLett.132.231002*
- Tohfa, H.; et al. (2023)."A Cosmic Microwave Background Search for Fine-Structure Constant Evolution." *Physical Review D (Vol. 109, No. 10). doi: 10.1103/PhysRevD.109.103529*
- Chatchadanoraset, J. , Tohfa, H. , including others (2026) "Breaking the Exoplanet Mass-Eccentricity Degeneracy: Combining Periodic Orbit Constraints with Transit Timing Variations" (In Review)
- Cain, C.; et al. including Tohfa, H. (2026)."Introducing SAGUARO – Simulating IGM Evolution and Environments At High Resolution: Setup and First Results." *Accepted to JCAP, preprint on ArXiv: 2603.25788*.
- Cain, C.; et al. including Tohfa, H. (2026)."A self-consistent analytical model for both the photoionization rate and reionization history." *Accepted to JCAP, preprint on ArXiv: 2606.19449*.
- Jacobson-Galan, W. V.; et al. including Tohfa, H. (2023)."SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova." *The Astrophysical Journal Letters, 954, L42. doi:10.3847/2041-8213/acf2ec*.

Awards and Recognition

American Institute of Physics SPS Outstanding Undergraduate Research Award, MIT Summer Research Grant, Velay Fellowship 2020, Bryn Mawr College Summer Science Research Grant, Bryn Mawr College Merit Scholarship 2018-2022

Research Experiences

Department of Astronomy, University of Washington September 2023 - Present
Advisor: Professor Andrew Connolly

- Developed AIDonut, the first neural network deployed for real-time control of telescope optics, now running in the Vera C. Rubin Observatory Active Optics System. Adapted a simulation-trained convolutional wavefront estimator to on-sky LSSTCam data across 3M donut stamps, testing five transfer learning strategies and redesigning the network with per-mode output heads and a bounded GradNorm loss to predict 21 Zernike coefficients plus seeing. Reduced wavefront estimation error by 20% relative to the forward-modeling baseline solver and cut inference from 30-40 seconds to under one second, meeting the 36-second visit cadence and shortening the correction latency from three visits to two (in prep).

- Built a convolutional surrogate for instrument signature removal on LSSTCam, replacing a nonlinear sequence of detector calibration steps with a single 282 ms evaluation. Reproduced fluxes to 0.1% against 0.7% for a linear gain-bias-flat model and reduced noise by 59% compared to a linear model, demonstrating recovery of nonlinear terms in the detector response. Extending to full-detector scale with crosstalk between amplifiers, toward an end-to-end path from raw amplifier data to wavefront correction within camera readout.

- Trained KBMOD, a moving target classifier for faint object detections usind ResNet-based classifier with squeeze-and-excitation channel attention, trained with focal loss on combined 2M sampled. Achieved 97% detection accuracy on held-out test set, recovering 3,318 novel high-confidence candidates not present in existing cross-match catalogs as potential trans-Neptunian objects (in prep).

Advisor: Professor Matthew McQuinn

- Developed deep learning emulator for predicting the ionizing photon mean free path during cosmic reionization, trained on high-resolution (2 kpc) radiation-hydrodynamic simulations spanning 126 parameter combinations. Built residual multi-layer perceptron achieving 1.6% median prediction error across four orders of magnitude in mean free path, enabling millisecond inference versus hundreds of CPU-hours per simulation. Applied emulator to constrain reionization history from observed mean free path measurements at z = 5-6, calculated ionizing emissivity that standard column density distribution evolution cannot explain (published).

- Developed a lossy neural compression method for cosmological simulation fields, quantizing the field on a fixed grid so reconstruction error is set by the science tolerance rather than by training, and learning only the bin probabilities with a two-scale autoregressive transformer. Benchmarked against a Gaussian coder and SZ3 across weak lensing maps, an N-body density field, and Lagrangian displacements, reaching percent-level accuracy at 1.3-5.3 bits per pixel versus 32 stored while preserving the power spectrum to better than 1%. Achieved the lowest rate on every field tested, 9-27% below the Gaussian coder and 6-21% below SZ3, measuring the structure that two-point statistics and local smoothness cannot represent (submitted, in review).

- Enhanced MP-Gadget simulation framework by implementing baryon streaming physics modules to analyze the impact of relative velocity between dark matter and baryons on halo mass distributions and formation dynamics.

Advisor: Professor Eric Agol

- Co-developed Julia package for detecting exoplanets using periodic orbit constraints and transit timing variations, breaking mass-eccentricity degeneracy that limits standard methods. Implemented Levenberg-Marquardt optimizer with Jacobians computed via automatic differentiation, achieving convergence in hundreds of iterations for high-dimensional parameter spaces. Validated by reproducing HR 8799's 4-planet periodic orbit (10,000-year stability), then applied to TRAPPIST-1 system revealing evidence for previously undetected planet (In review)

- **Co-I on JWST Cycle 4 Program #12492** (accepted) to confirm candidate planet through follow-up observations

Center for Astrophysics, Harvard University August 2026 - Present
Collaborator: Dr Mike Smith

- Developing multi scale foundation model based on how information cluster across different physical scales.

Center for computational astrophysics, Flatiron Institute April 2026 - Present
Advisor: Dr Francisco Villaescusa-Navarro

- Developed a semantic figure retrieval system for astronomy literature, using dense embeddings (bge-base-en-v1.5), vector similarity search, and vision-language model verification. Built a caption-based index over 482,750 figures from 53,839 arXiv papers, and designed a multi-register benchmark to evaluate retrieval quality and outperformed existing literature search baselines across all query types (Accepted)

Department of physics, UC Riverside September 2022 - August 2023
Advisor: Professor Simeon Bird

- Implemented wavelet scattering transforms as feature extraction technique, demonstrating order-of-magnitude improvement in parameter constraint precision over traditional power spectrum methods using Fisher information matrix analysis (Published).

Microsoft Research, Redmond, WA
Research Intern May 2022 - August 2022
Mentor: Dr. Simon David William Frost

- Designed noise filtering pipeline and feature extraction system for multi-class classification problems. Developed time-series LSTM network that outperformed existing GAN baseline by 10%, implementing comprehensive evaluation framework to validate model performance against ecological ground truth data.

Department of Physics and Astronomy, Haverford College
Advisor: Professor Daniel Grin

- Numerically solved dynamic equations of scalar field for BSBM and Runaway Dilaton model, analytically solving the equations at boundaries to validate results, constrained free parameters in the models with Planck 2018, QSO data using Principal Component Analysis and Monte Carlo simulation, forecasted uncertainties in the possible detection of the fine structure constant in Simons Observatory CMB experiment (Published).

Kavli Institute for Astrophysics, Massachusetts Institute of Technology
Research Intern May 2021 - September 2021
Advisor: Professor Michael McDonald

- Numerically showed that relationship between the pressure of X-Ray cavities and luminosity can be generalized as a cluster property and analytically proving that the geometry of the cavity is the main catalyst behind the scatter seen in the pressure-luminosity relationship, determed the exact correlation between distance from the cavity and cavity radius where most of the detected cavities are seen.

Department of Physics, Bryn Mawr College
Undergraduate Research Assistant May 2019 - August 2019
Advisor: Professor Michael Schulz

- Analytically solved the field equations for Lifshitz metric and finding the correlation function using AdS/CFT correspondence that specified ultralocality of the field.

Department of Physics, Bryn Mawr College
Undergraduate Research Assistant December 2018 - May 2019
Advisor: Professor David Schaffner
Wrote a program in Python using bi-spectral analysis to find the bi-coherence to see at which energy modes the energy transfer takes place in plasma. The program is now being used to understand the stability of transient spiral arms in galaxies.
Relevant Course Work: Machine Learning, Statistical Methods, Computational Physics, Data Structures and Algorithms, Applied Statistics, Advanced Mathematics (Real Analysis, Linear Algebra, Differential Equations)

Presentations

Tohfa. Hurum Maksora, "AIDonut : First Real-Time Neural Network Control of Telescope Active Optics the Vera Rubin Observatory," The 2026 AstroAI Workshop, Harvard University, 2026
Tohfa. Hurum Maksora, "From Simulations to Emulators: Improving IGM Opacity Models for Cosmic Reionization," Kaba Kada: Exploring the First Billion Years of the Universe, Queensland, Australia, 2025
Tohfa. Hurum Maksora, "AI methods for Rubin Active Optics System", Rubin Community Workshop, Stanford University, 2026
Tohfa. Hurum Maksora, "Machine Learning in Modern Astronomy **(Invited Speaker)**, Haverford College, 2026
Tohfa. Hurum Maksora, "Forecast Cosmological Constraints with the 1D Wavelet Scattering Transform and the Lyman-alpha forest", University of Washington, 2023
Tohfa. Hurum Maksora, "Fine structure constant variation," American Physical Society, NY, 2022
Tohfa. Hurum Maksora, "Fine structure constant variation," American Astronomical Society, CA, 2022
Tohfa. Hurum Maksora, "Understanding X-ray Cavities," MIT Summer Research Symposium, Massachusetts Institute of Technology, MA, 2021
Tohfa. Hurum, Pandey, Shiksha, Khan, Faryal, "Computing Lifshitz Field Theory Correlation Functions Using the AdS/CFT Correspondence," Summer Science Research Symposium, Bryn Mawr College, PA, 2019
Tohfa. Hurum, Pandey, Shiksha, "Bispectrum Analysis on non-linear system," Spring Research Symposium, Bryn Mawr College, PA, 2019

Workshops:

International High Performance Computing Summer School 2026, Perth, Asutralia; Rubin DP2 Sprint; LICK workshop in observational astronomy 2023; Center for Matter at Atomic Pressures Summer School 2021 (selected to receive a stipend); CMB-S4 Summer School 2021

Teaching Experience

University of Washington, Seattle, WA
Teaching Assistant for Introduction to Astronomy (Fall 2023)
Teaching Assistant for The Planets (Winter 2024)
Teaching Assistant for The Planets (Spring 2024)
Haverford College, Haverford, PA
Teaching Assistant for Phys-309 (Advanced Electromagnetism), Fall 2021
Teaching Assistant for Phys-303 (Statistical Mechanics and Thermodynamics), Fall 2021
Teaching Assistant for Phys-302 (Advanced Quantum Mechanics), Spring 2021
Teaching Assistant for Phys-308 (Advanced Classical Mechanics), Fall 2020
Teaching Assistant for Math-121 (Multivariable Calculus), Spring 2020
Student Consultant for Teaching and Learning Center (Fall 2019, Summer 2020, Fall 2020, Fall 2021)
Bryn Mawr College, Bryn Mawr, PA
Peer Tutor (Introductory Physics, Modern Physics), Fall 2020 – Spring 2022
Quantitative Center Tutor, Fall 2020 – Spring 2022
Teaching Assistant: Introductory Physics I (Fall 2019) and Phys-102: Introductory Physics II (Spring 2019 and Spring 2020)

Outreach

Mentoring: Micah J. Kaczynski (Pre-MAP, University of Washington)
Volunteering: UW Planetarium - presented shows for K–12 school groups and public Friday-night audiences; ESTEAM - President of a youth prison education program, serving as tutor and running program administration
Organizing: Rubin Collaboration Meeting (session chair); UW AstroFest (session chair)